

Name: _____



Problem solving with standard form

Task A: Not quite standard form numbers

1) Write the following in correct standard form:

- a) 398×10^{34}
- b) 9000×10^{24}
- c) 340×10^9
- d) $89\,000 \times 10^{18}$
- e) 42×10^{-5}
- f) $490\,000 \times 10^{-12}$
- g) 99×10^{-2}

2) Write the following in correct standard form:

- a) 0.034×10^{39}
- b) 0.0056×10^{23}
- c) $0.000\,032 \times 10^9$
- d) 0.45×10^5
- e) 0.583×10^{-2}
- f) $0.003\,53 \times 10^{-3}$
- g) 0.1×10^1

3) Write the following in correct standard form:

- a) 3590×10^9
- b) $0.003\,59 \times 10^9$
- c) 35.9×10^{-9}
- d) 0.359×10^{-9}
- e) 359×10^{-2}
- f) 359×10^2
- g) 35.9×10^{-2}



Task B: Using standard form in problems

1) How many years are 1 billion seconds?

Use a calculator and use 365 days as a standard year.

2) Flash drives come in 16 MB, 64 MB and 128 MB. A video was made and has 21 700 000 bytes.

Given that 1×10^6 is 1 MB (megabyte).

What is the minimum flash drive needed to put this film on?

3) The radius of Mercury is approximately 2 500 km.

The radius of Neptune is approximately 25 000 000 m.

How many times bigger is Neptune than Mercury?

Planet	$\approx$ Radius
Mercury	2 500 km
Venus	6 000 km
Earth	6 000 km
Mars	3 000 km
Jupiter	71 000 km
Saturn	60 000 km
Uranus	25 000 km
Neptune	25 000 km

4) Here are some approximations of the radii of planets.

Write any two of your own questions, including the use of standard form.

